

MBA Fastrack 2025 (CAT + OMETs)

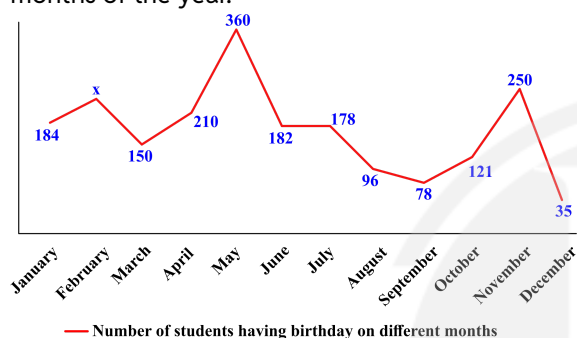
Data Interpretation & Logical Reasoning

DPP: 5

Multiple Charts

Directions (1-5) Read the following passage and answer the given questions.

There are 2100 students in a college and they all have their birthday in one of the twelve months of a year. The line graph given below shows the number of students having birthdays in different months of the year.



The table given below shows the ratio of the number of boys to girls having birthdays in the first six months of a year

Month	Ratio of number of boys to number of girls
January	13: 10
February	7: 9
March	3: 2
April	4: 3
May	7: 11
June	9: 5

Q1 Find the difference between the numbers of boys who have birthdays in January to that of February.

- (A) 6 (B) 8
(C) 12 (D) 14

Q2 Find the average number of girls who have birthdays in January, March, April, May and June.

- (A) 97 (B) 93
(C) 107 (D) 103

Q3 If the average money spent on a birthday party by each of the students who have

birthdays in September and October is Rs. 48 and Rs. 22, respectively, then find the difference between the total money spent on a party by all the students who have birthdays in September and that of October.

- (A) Rs 1082 (B) Rs 1162
(C) RS 964 (D) Rs 1246

Q4 If the ratio of the number of boys to girls who have birthdays in November and December is 13:12 and 2:3, respectively, then find the total number of girls who have birthdays in November and December combined.

- (A) 145 (B) 139
(C) 151 (D) None of these

Q5 If 25% of the students who have birthdays in August are girls, and out of these girls, 7 are from the Science department, then find the number of girls from other departments than Science who have birthdays in August.

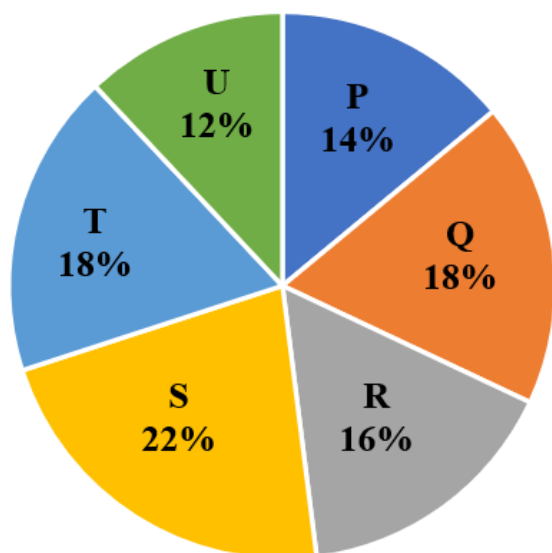
- (A) 15 (B) 16
(C) 17 (D) 18

Directions (6-10) Read the following passage and answer the given questions.

The pie chart below shows the percentage breakup of students enrolling in various colleges during 2015.

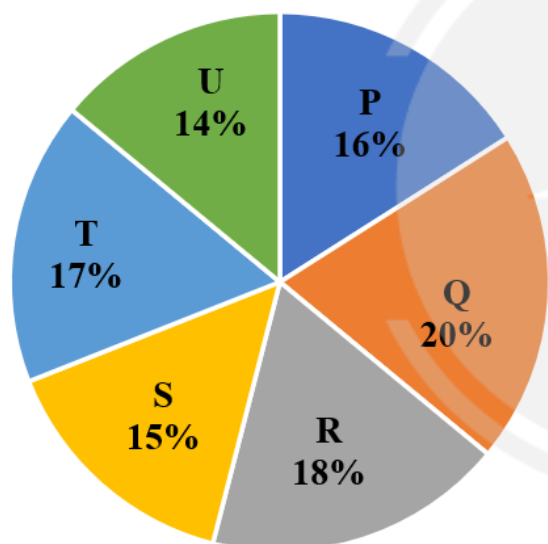
Total students = 52500 (Male + Female)





The pie chart below shows the percentage breakup of female students enrolling in various colleges during 2015.

Total female students = 24000



Q6 What is the ratio of male students enrolled in college P and T together to female students enrolled in college R and U taken together?

- (A) 30 : 23 (B) 74 : 57
(C) 15 : 13 (D) 37 : 32

Q7 If the ratio of BSC, BCOM and BA students enrolled in college R is 4 : 5 : 6. Then find the total number of students enrolled for BCOM in college R (Assume only these three branches are there in College R)?

- (A) 2500 (B) 2600

(C) 2700

(D) 2800

Q8 Female students from S and T taken together are approximately what percent less than male students from Q and T taken together?

- (A) 29.4% (B) 23.5%
(C) 20.4% (D) 25.5%

Q9 Male and female students from college Q during 2016 increased by 20% and 25% respectively then, what is the percentage change in UG students from Q during 2015 to 2016 if the ratio of UG to PG students during 2015 and 2016 is 8 : 7 and 3 : 2 respectively?

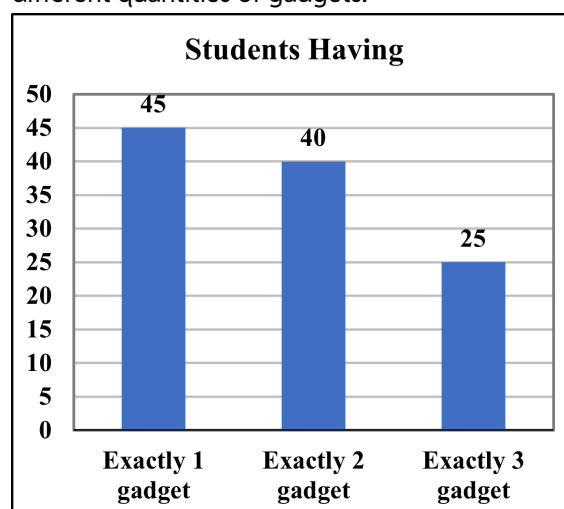
- (A) 37.8% (B) 34.4%
(C) 22.5% (D) 35.4%

Q10 If 80%, 75%, 80% and 90% of students from colleges Q, R, S and U respectively passed in the examination then the approximate average students passing from these 4 colleges are?

- (A) 7181 (B) 7192
(C) 7103 (D) 7114

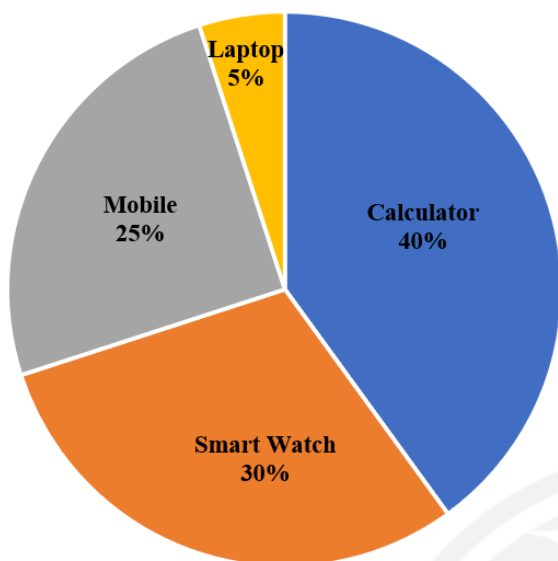
Directions (11-15) Read the following passage and answer the given questions.

In a class of exactly 120 students, each student possesses no more than three gadgets. These gadgets include Smartwatches, Mobiles, Laptops, and Calculators, with no other gadgets used. The bar graph displays the number of students with different quantities of gadgets.



The accompanying pie chart illustrates the distribution of the number of gadgets each student owns.

Number of Gadgets



- Q11** How many students have none of the given gadgets?
- Q12** Find the ratio of the students having exactly 1 gadget to the total number of laptops?
 (A) 3:4 (B) 5:4
 (C) 9:2 (D) None of these
- Q13** All the calculators are equally distributed only among the students who have exactly 1 gadget and exactly 2 gadgets, then minimum how many students with exactly 1 gadget do not have calculators?
 (A) 10 (B) 15
 (C) 0 (D) 5
- Q14** If no students have more than one gadget of the same type, then at least how many calculators are used by the students who are using exactly 2 gadgets?
 (A) 10 (B) 15
 (C) 5 (D) 0
- Q15** If all Smart Watches are owned by students who have exactly 3 gadgets each, what is the maximum number of these students who have only smartwatches?
 (A) 12 (B) 20

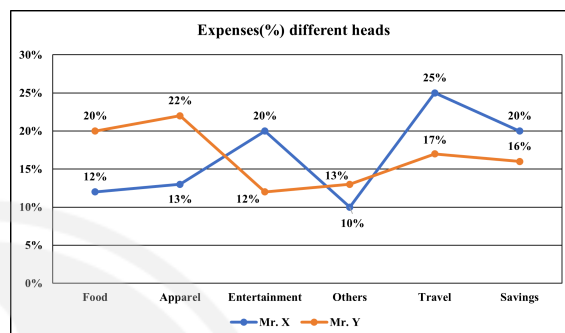
(C) 18

(D) 22

Directions (16-20) Read the following passage and answer the given questions.

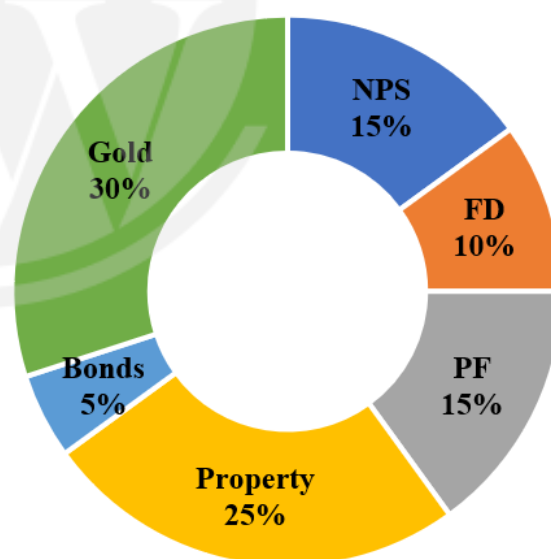
The Shah family has two earning members, Mr. X Shah and Mrs. Y Shah. Both are taking the family's expenses of different heads. Both Mr. X and Mrs. Y have combined savings, not necessarily equal.

The line chart shows the distribution of the total income of Mr. X and Mrs. Y in different heads.



The doughnut chart below shows the percentage distribution of their part of the combined savings in the different portfolios.

Savings (%)



Note: Savings have to be considered apart from expenses.

- Q16** If 10% of Mrs. Y's income is equal to the amount invested in PF, what is the income ratio between Mr. X and Mrs. Y?
 (A) 25:27 (B) 27:19
 (C) 15:23 (D) 38:15



Q17 If the amount invested in Property is the same as the amount that Mrs. Y spent on Travel, then what is the percentage of the expenses of Mr. X to that of expenses of Mrs. Y?

- (A) 153.65% (B) 247.62%
(C) 268.72% (D) None of these

Q18 If the ratio of savings of Mr. X and Mrs. Y is 4:5, then what is the relation between the total income of Mr. X to that of Mrs. Y?

- (A) Mr. X > Mrs. Y
(B) Mr. X = Mrs. Y
(C) Mr. X < Mrs. Y
(D) Can not be determined

Q19 If Mrs. Y's earnings exceed Mr. X by 50%, find in how many heads the expenses of Mrs. Y is/ are lower than Mr. X's?

- (A) 4 (B) 1
(C) 2 (D) 3

Q20 If the amount saved by Mr. X is less than that by Mrs. Y, then under how many heads is the amount spent by Mrs. Y definitely more than that by Mr. X?

- (A) 4 (B) 3
(C) 2 (D) 1

Directions (21-25) Read the following passage and answer the given questions.

At a toll plaza there are five lanes P, Q, R, S, and T at which vehicles can pass through the plaza.

Note 1: Vehicles passed are either 2 wheelers or 4 wheelers.

Note 2: Toll tax for each 2 - wheelers vehicle is Rs.180 and toll tax for each 4 - wheelers vehicle is Rs.240.

Pie chart given below shows the percent distribution of total vehicles passed from the toll plaza at different lanes on a particular day.

Total vehicles passed from different lanes of the toll plaza

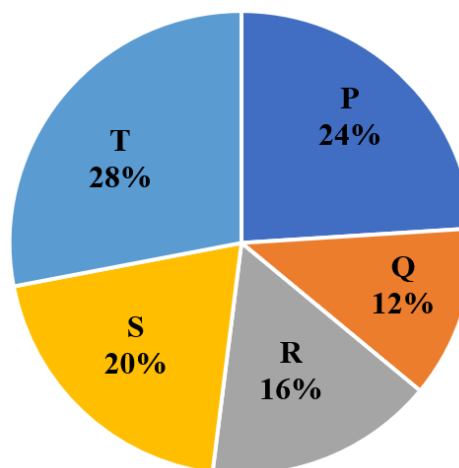


Table given below shows the ratio of total revenue collected from 2 - wheelers to the total revenue collected from 4 - wheelers at different lanes of the toll plaza.

Lanes	Ratio of total revenue collected from 2- wheelers to that from 4-wheelers
P	3:2
Q	15:16
R	21:20
S	3:2
T	39:32

Note: Total revenue collected at toll plaza from all the five lanes together on that day is Rs.15180.

Q21 Ratio of total wheels of all the vehicles passed from lane P to lane S of the toll plaza is $(2a - 2) : (a + 3)$, then which of the following is not TRUE?

- (A) $\sqrt[3]{a+1} = 2$
(B) $\sqrt{a+2} = 3$
(C) $a - 3 = 4$
(D) $(a + 8) \div 3 = 6$

Q22 What is the difference between total 2 - wheelers and total 4 - wheelers passed from the toll plaza though all the five lanes together?

- (A) 21 (B) 19
(C) 23 (D) 17

Q23 Consider the following statements given below and mark the correct option.

x = Difference between total 2 - wheelers and 4 - wheelers passed from lane S.

y = Total 4 - wheelers passed from lane Q + 1



- (A) $x > y$
 (B) $x + y = 8$
 (C) $x^2 + y^2 = 25$
 (D) $x - y = 0$

Q24 Total revenue collected from all the vehicles from lane R of the toll plaza is what percent of total revenue collected from all the vehicles from lane S of that toll plaza?

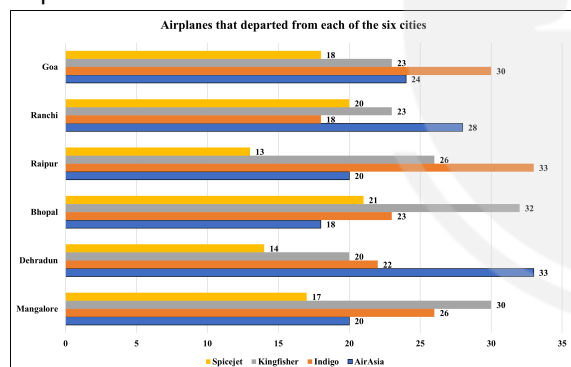
- (A) 74% (B) 88%
 (C) 78% (D) 82%

Q25 What is the sum of the number of 2 wheeler passed from lane P, R and S?

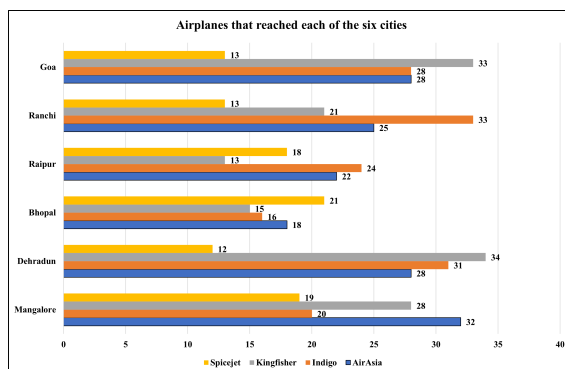
Directions (26-30) Read the following passage and answer the given questions.

The bar graphs below present information regarding the aircraft of four airlines—AirAsia, Indigo, Kingfisher, and SpiceJet—operating between six distinct cities over one week.

Graph 1 illustrates the number of airplanes that departed from each of the six cities.



Graph 2 depicts the number of airplanes that arrived at each of the six cities.



Throughout the specified week, each company operated at least one airplane from each of the six cities to each of the other five cities (excluding the departure city).

Note: Assume the airplanes of these four airlines departed from and arrived at all six cities during the week.

Q26 What is the maximum number of AirAsia airplanes that departed from Mangalore and reached Bhopal?

Q27 What is the maximum number of Kingfisher airplanes that departed from Bhopal and reached Raipur?

Q28 What is the maximum number of airplanes that departed from Goa and reached Dehradun?

Q29 What is the maximum number of SpiceJet airplanes that departed from any one of the given cities to any other five cities?

Q30 The maximum number of AirAsia and Indigo airplanes that departed from Raipur and reached Goa?



Answer Key

Q1 (B)
Q2 (D)
Q3 (A)
Q4 (D)
Q5 (C)
Q6 (D)
Q7 (D)
Q8 (B)
Q9 (A)
Q10 (B)
Q11 10
Q12 (C)
Q13 (C)
Q14 (A)
Q15 (B)

Q16 (D)
Q17 (B)
Q18 (C)
Q19 (B)
Q20 (B)
Q21 (D)
Q22 (B)
Q23 (D)
Q24 (D)
Q25 29
Q26 14
Q27 9
Q28 73
Q29 16
Q30 40



Hints & Solutions

Note: scan the QR code to watch video solution

Q1. Text Solution:

Total number of students in the college = $184 + x + 150 + 210 + 360 + 182 + 178 + 96 + 78 + 121 + 250 + 35$

$184 + x + 150 + 210 + 360 + 182 + 178 + 96 + 78 + 121 + 250 + 35 = 2100$

$x = 256$

Month	Number of boys	Number of girls
January	$\frac{13}{23} \times 184 = 104$	$\frac{10}{23} \times 184 = 80$
February	$\frac{7}{16} \times 256 = 112$	$\frac{9}{16} \times 256 = 144$
March	$\frac{3}{5} \times 150 = 90$	$\frac{2}{5} \times 150 = 60$
April	$\frac{4}{7} \times 210 = 120$	$\frac{3}{7} \times 210 = 90$
May	$\frac{7}{18} \times 360 = 140$	$\frac{11}{18} \times 360 = 220$
June	$\frac{9}{14} \times 182 = 117$	$\frac{5}{14} \times 182 = 65$

Required difference = $112 - 104 = 8$.

Video Solution:



Q2. Text Solution:

Month	Number of boys	Number of girls
January	$\frac{13}{23} \times 184 = 104$	$\frac{10}{23} \times 184 = 80$
February	$\frac{7}{16} \times 256 = 112$	$\frac{9}{16} \times 256 = 144$
March	$\frac{3}{5} \times 150 = 90$	$\frac{2}{5} \times 150 = 60$
April	$\frac{4}{7} \times 210 = 120$	$\frac{3}{7} \times 210 = 90$
May	$\frac{7}{18} \times 360 = 140$	$\frac{11}{18} \times 360 = 220$
June	$\frac{9}{14} \times 182 = 117$	$\frac{5}{14} \times 182 = 65$

Required average = $\frac{80+60+90+220+65}{5} = 103$

Video Solution:



Q3. Text Solution:

Month	Number of boys	Number of girls
January	$\frac{13}{23} \times 184 = 104$	$\frac{10}{23} \times 184 = 80$
February	$\frac{7}{16} \times 256 = 112$	$\frac{9}{16} \times 256 = 144$
March	$\frac{3}{5} \times 150 = 90$	$\frac{2}{5} \times 150 = 60$
April	$\frac{4}{7} \times 210 = 120$	$\frac{3}{7} \times 210 = 90$
May	$\frac{7}{18} \times 360 = 140$	$\frac{11}{18} \times 360 = 220$
June	$\frac{9}{14} \times 182 = 117$	$\frac{5}{14} \times 182 = 65$

Required difference = $(78 \times 48) - (121 \times 22)$

$= 3744 - 2662$

$= \text{Rs. } 1082$.

Video Solution:



Q4. Text Solution:

Month	Number of boys	Number of girls
January	$\frac{13}{23} \times 184 = 104$	$\frac{10}{23} \times 184 = 80$
February	$\frac{7}{16} \times 256 = 112$	$\frac{9}{16} \times 256 = 144$
March	$\frac{3}{5} \times 150 = 90$	$\frac{2}{5} \times 150 = 60$
April	$\frac{4}{7} \times 210 = 120$	$\frac{3}{7} \times 210 = 90$
May	$\frac{7}{18} \times 360 = 140$	$\frac{11}{18} \times 360 = 220$
June	$\frac{9}{14} \times 182 = 117$	$\frac{5}{14} \times 182 = 65$

Required number of girls

$= \frac{12}{25} \times 250 + \frac{3}{5} \times 35 = 120 + 21 = 141$.

Video Solution:



Q5. Text Solution:



Month	Number of boys	Number of girls
January	$\frac{13}{23} \times 184 = 104$	$\frac{10}{23} \times 184 = 80$
February	$\frac{7}{16} \times 256 = 112$	$\frac{9}{16} \times 256 = 144$
March	$\frac{3}{5} \times 150 = 90$	$\frac{2}{5} \times 150 = 60$
April	$\frac{4}{7} \times 210 = 120$	$\frac{3}{7} \times 210 = 90$
May	$\frac{7}{18} \times 360 = 140$	$\frac{11}{18} \times 360 = 220$
June	$\frac{9}{14} \times 182 = 117$	$\frac{5}{14} \times 182 = 65$

Required number of girls from other departments than Science who have birthday on August = (25% of 96) – 7

$$= 24 - 7$$

$$= 17.$$

Video Solution:



Q6. Text Solution:

Students enrolled in various colleges during 2015 can be calculated as:

From P:

Enrolled = 7350

Female = 3840

Male = (7350 – 3840) = 3510

Based on the above data we get:

College	Enrolled	Female	Male
P	7350	3840	3510
Q	9450	4800	4650
R	8400	4320	4080
S	11550	3600	7950
T	9450	4080	5370
U	6300	3360	2940
Total	52500	24000	28500

Male students enrolled in college P and T together = (3510 + 5370) = 8880

Female students enrolled in college R and U together = (4320 + 3360) = 7680

Required ratio = 8880 : 7680 = 37 : 32

Video Solution:



Q7. Text Solution:

College	Enrolled	Female	Male
P	7350	3840	3510
Q	9450	4800	4650
R	8400	4320	4080
S	11550	3600	7950
T	9450	4080	5370
U	6300	3360	2940
Total	52500	24000	28500

Students enrolled for BCOM in college R =

$$8400 \times \frac{5}{15}$$

$$= 2800$$

Video Solution:



Q8. Text Solution:

College	Enrolled	Female	Male
P	7350	3840	3510
Q	9450	4800	4650
R	8400	4320	4080
S	11550	3600	7950
T	9450	4080	5370
U	6300	3360	2940
Total	52500	24000	28500

Female students from S and T = (3600 + 4080) = 7680

Male students from Q and T = (4650 + 5370) = 10020

$$\text{Required percent} = \frac{10020 - 7680}{10020} \times 100$$

$$= 23.35\%$$

$$= 23.5\% \text{ approx.}$$



Video Solution:



Q9. Text Solution:

College	Enrolled	Female	Male
P	7350	3840	3510
Q	9450	4800	4650
R	8400	4320	4080
S	11550	3600	7950
T	9450	4080	5370
U	6300	3360	2940
Total	52500	24000	28500

Male students from Q during 2016 = $4650 \times \frac{120}{100}$
= 5580

Female students from Q during 2016 =
 $4800 \times \frac{125}{100} = 6000$

Total students from Q during 2016 = (5580 +
6000) = 11580

UG students from Q during 2016 = $11580 \times \frac{3}{5} =$
6948

UG students from Q during 2015 = $9450 \times \frac{8}{15}$
= 5040

Required percent = $\frac{6948 - 5040}{5040} \times 100$
= 37.8% (approx.)

Video Solution:



Q10. Text Solution:

College	Enrolled	Female	Male
P	7350	3840	3510
Q	9450	4800	4650
R	8400	4320	4080
S	11550	3600	7950

T	9450	4080	5370
U	6300	3360	2940
Total	52500	24000	28500

Passed students from college Q = 80% of 9450 =
7560

Passed students from college R = 75% of 8400 =
6300

Passed students from college S = 80% of 11550 =
9240

Passed students from college U = 90% of 6300 =
5670

Required average = $\frac{7560 + 6300 + 9240 + 5670}{4}$
= 7192 (approx.)

Video Solution:



Q11. Text Solution:

The following table shows the data for number of
students and the total number of gadgets a
student have:

	Students having	Total
Exactly 1 gadget	45	$45 \times 1 = 45$
Exactly 2 gadgets	40	$40 \times 2 = 80$
Exactly 3 gadgets	25	$25 \times 3 = 75$
Total	$45 + 40 + 25$ = 110	$45 + 80 + 75$ = 200

Hence, the number of students having none of
the given gadgets = $120 - 110 = 10$

The total number of gadgets students have =
200

Given, 40% of the gadgets are calculator = 40%
of 200 = 80

Similarly, the following table shows the
distribution of the number of gadgets students
have: -

Distribution of (200)



number of gadgets	In %	number
Calculator	40%	80
Smart watch	30%	60
Mobile	25%	50
Laptop	5%	10

10 students have none of the given gadgets.

Video Solution:



Q12. Text Solution:

the following table shows the distribution of the number of gadgets students have: -

Distribution of (200)		
number of gadgets	In %	number
Calculator	40%	80
Smart watch	30%	60
Mobile	25%	50
Laptop	5%	10

Students having exactly 1 gadget = 45

Total number of laptops = 10

Required ratio = 45:10 = 9:2

Video Solution:



Q13. Text Solution:

the following table shows the distribution of the number of gadgets students have: -

Distribution of (200)		
-----------------------	--	--

number of gadgets	In %	number
Calculator	40%	80
Smart watch	30%	60
Mobile	25%	50
Laptop	5%	10

the number of calculators = 80

To minimize the number of students with exactly 41 gadgets who do not have calculator

The students who have exactly 1 gadget who will get calculator = 45

The students who have exactly 2 gadget who will get calculator = 35

So, the minimum number of students with exactly 1 gadget that does not have calculator = 0

Video Solution:



Q14. Text Solution:

the number of calculators is = 80

For at least using exactly 2 gadgets, there must be at the most number of calculators using both exactly 1 and exactly 3 gadgets.

Hence, the at the most number of calculators using both exactly 1 and exactly 3 gadgets = 45 + 25 = 70

So, the at least number of calculators using exactly 2 gadgets = 80 - 70 = 10.

Video Solution:



Q15. Text Solution:



the following table shows the distribution of the number of gadgets students have: -

Distribution of (200)		
number of gadgets	In %	number
Calculator	40%	80
Smart watch	30%	60
Mobile	25%	50
Laptop	5%	10

The number of smartwatches = 60

The maximum number of these students who have only smartwatches

So, these 60 smartwatches are with $\frac{60}{3} = 20$ students.

Video Solution:



Q16. Text Solution:

Data from the Line chart: -

Expenses	Mr. X	Mrs. Y
Food	12%	20%
Apparel	13%	22%
Entertainment	20%	12%
Others	10%	13%
Travel	25%	17%
Savings	20%	16%
Total	100%	100%

Data from the Doughnut chart: -

Portfolios	Savings (%)
NPS	15%
FD	10%
PF	15%
Property	25%
Bonds	5%
Gold	30%
Total	100%

Let total savings be $100x$.

As PF is 15% of the total savings,

PF = $15\%(100x) = 15x$, which is the same as 10% of Mrs. Y's income.

Total income Mrs. Y's = $(15x/10)100 = 150x$

Hence Mrs. Y's income is $150x$.

Savings of Mrs. Y = $16\%(150x) = 24x$

So, the savings of Mr. X = Total savings – Mrs. Y's savings

the savings of Mr. X = $100x - 24x$

the savings of Mr. X = $76x$, which is the same as 20% of Mr. X's income.

Mr. X's income = $(76x/20)100 = 380x$

Hence, the ratio of the incomes of Mr. X and Mrs. Y = $380x:150x = 38:15$.

Video Solution:



Q17. Text Solution:

Let total savings be $100x$.

As Property forms 25% of savings,

Property = $25\%(100x) = 25x$ which is the same as the amount that Mrs. Y spent on Travel

Mrs. Y spent on Travel is = 17% of the Mrs. Y's income = $25x$

Expenses of Mrs. Y = 84% of the Mrs. Y's income.

Expenses of Mrs. Y = $\frac{25x}{17} \times 84 = 123.53x$

Savings of Mrs. Y = $\frac{25x}{17} \times 16 = 23.53x$

So, the savings of Mr. X = Total savings – Mrs. Y's savings

the savings of Mr. X = $100x - 23.53x$

the savings of Mr. X = $76.47x$, which is the same as 20% of Mr. X's income.

So Mr. X's income = $\frac{76.47x}{20} \times 80 = 305.88x$

Hence, the required percentage = $\frac{305.88x}{123.53x} \times 100 = 247.62\%$

Video Solution:



**Q18. Text Solution:**

It is given, the ratio of savings of Mr. X and Mrs. Y is 4:5

Let the savings of, Mr. X = $4x$, and Mrs. Y = $5x$

The savings of Mr. X is 20% of his income,

So, the income of Mr. X = $\frac{4x}{20} \times 100 = 20x$

The savings of Mrs. Y is 16% of her income,

So, the income of Mrs. Y = $\frac{5x}{16} \times 100 = 31.25x$

Hence, the income of Mr. X is less than that of Mrs. Y.

Video Solution:**Q19. Text Solution:**

Considering Mr. X's income is $100x$, then Mrs. Y's income will be $150x$.

The below table gives the details of the expenses in different heads.

Heads of expense	Mr. X = $100x$	Mr. X's Expense	Mrs. Y = $150x$	Mrs. Y's Expense
Food	12%	$12x$	20%	$30x$
Apparel	13%	$13x$	22%	$33x$
Entertainment	20%	$20x$	12%	$18x$
Others	10%	$10x$	13%	$19.5x$
Travel	25%	$25x$	17%	$25.5x$

So, only in entertainment, the expenses of Mrs. Y are lower than Mr. X's.

Video Solution:**Q20. Text Solution:**

Let the amount saved by Mrs. Y be $16x$. So, the total savings of Mrs. Y is $100x$

As per the question, let Mr. X's savings be $15.9y$.

Hence Mr. X's savings will be $\frac{15.9y}{20} \times 100 = 79.5y$

The breakup of expenses for both:

Heads of expense	Mr. X = $79.5y$	Mr. X's Expense	Mrs. Y = $100x$	Mrs. Y's Expense
Food	12%	$9.54y$	20%	$20x$
Apparel	13%	$10.335y$	22%	$22x$
Entertainment	20%	$15.9y$	12%	$12x$
Others	10%	$7.95y$	13%	$13x$
Travel	25%	$19.875y$	17%	$17x$
Savings	20%	$15.9y$	16%	$16x$
Total		$79.5y$		$100x$

In Food, Apparel, and Others head of expenses, Mrs. Y's expenses are definitely more than Mr. X's.

Video Solution:**Q21. Text Solution:**

Let the total vehicles passed from lanes

P, Q, R, S , and T of the toll plaza is ' $24x$ ', ' $12x$ ', ' $16x$ ', ' $20x$ ', and ' $28x$ ' respectively.

Total revenue collected = Total vehicles $\times$ Tax per vehicles

Total vehicle = $\frac{\text{Total revenue collected}}{\text{Total tax per vehicle}}$

Hence, the ratio of total 2-wheelers to 4-wheelers from a lane is equal to the total revenue collected divided from 2-wheelers to 4-wheelers divided by respective toll tax per vehicle.

Lane P:

Ratio of 2-wheelers to 4-wheelers passed

$$= \left(\frac{3}{180}\right) : \left(\frac{2}{240}\right) = 2 : 1$$

$$2\text{-wheelers passed} = 24x \times \left(\frac{2}{3}\right) = 16x$$

$$4\text{-wheelers passed} = 24x \times \left(\frac{1}{3}\right) = 8x$$

Lane Q:

Ratio of 2-wheelers to 4-wheelers passed

$$= \left(\frac{15}{180}\right) : \left(\frac{16}{240}\right) = 5 : 4$$



$$2\text{-wheelers passed} = 12x \times \left(\frac{5}{9}\right) = \frac{20x}{3}$$

$$4\text{-wheelers passed} = 12x \times \left(\frac{4}{9}\right) = \frac{16x}{3}$$

Lane R:

$$\text{Ratio of 2-wheelers to 4-wheelers passed} = \left(\frac{21}{180}\right) : \left(\frac{20}{240}\right) = 7 : 5$$

$$2\text{-wheelers passed} = 16x \times \left(\frac{7}{12}\right) = \frac{28x}{3}$$

$$4\text{-wheelers passed} = 16x \times \left(\frac{5}{12}\right) = \frac{20x}{3}$$

Lane S:

$$\text{Ratio of 2-wheelers to 4-wheelers passed} = \left(\frac{3}{180}\right) : \left(\frac{2}{240}\right) = 2 : 1$$

$$2\text{-wheelers passed} = 20x \times \left(\frac{2}{3}\right) = \frac{40x}{3}$$

$$4\text{-wheelers passed} = 20x \times \left(\frac{1}{3}\right) = \frac{20x}{3}$$

Lane T:

$$\text{Ratio of 2-wheelers to 4-wheelers passed} = \left(\frac{39}{180}\right) : \left(\frac{32}{240}\right) = 13 : 8$$

$$2\text{-wheelers passed} = 28x \times \left(\frac{13}{21}\right) = \frac{52x}{3}$$

$$4\text{-wheelers passed} = 28x \times \left(\frac{8}{21}\right) = \frac{32x}{3}$$

Total 2-wheelers passed from the toll plaza =

$$16x + \left(\frac{20x}{3}\right) + \left(\frac{28x}{3}\right) + \left(\frac{40x}{3}\right) + \left(\frac{52x}{3}\right) = \frac{(48 + 20 + 28 + 40 + 52)x}{3} = \frac{188x}{3}$$

Total 4-wheelers passed from the toll plaza =

$$8x + \left(\frac{16x}{3}\right) + \left(\frac{20x}{3}\right) + \left(\frac{20x}{3}\right) + \left(\frac{32x}{3}\right) = \frac{(24 + 16 + 20 + 20 + 32)x}{3} = \frac{112x}{3}$$

$$\begin{aligned} \text{Total revenue collected} &= \left[\left(\frac{188x}{3}\right) \times 180 \right] + \left[\left(\frac{112x}{3}\right) \times 240 \right] \\ &= 15180 \end{aligned}$$

$$188x \times 60 + 112x \times 80 = 15180$$

$$188x \times 6 + 112x \times 8 = 1518$$

$$1128x + 896x = 1518$$

$$2024x = 1518$$

$$x = \frac{1518}{2024} = \frac{(2 \times 3 \times 11 \times 23)}{(2 \times 2 \times 2 \times 11 \times 23)}$$

$$x = \frac{3}{4}$$

Lanes	Total 2-wheelers passed	Total 4-wheelers passed
P	$16x = 12$	$8x = 6$
Q	$\frac{20x}{3} = 5$	$\frac{16x}{3} = 4$
R	$\frac{28x}{3} = 7$	$\frac{20x}{3} = 5$
S	$\frac{40x}{3} = 10$	$\frac{20x}{3} = 5$
T	$\frac{52x}{3} = 13$	$\frac{32x}{3} = 8$

Total wheels of all the vehicles passed from lane

P

$$= (12 \times 2) + (6 \times 4) = 24 + 24 = 48$$

Total wheels of all the vehicles passed from lane

S

$$= (10 \times 2) + (5 \times 4) = 20 + 20 = 40$$

According to the question:

$$48 : 40 = (2a - 2) : (a + 3)$$

$$6 : 5 = (2a - 2) : (a + 3)$$

$$6(a + 3) = 5(2a - 2)$$

$$6a + 18 = 10a - 10$$

$$4a = 28$$

$$a = 7$$

Option (a):

$$= \sqrt[3]{a + 1} = \sqrt[3]{7 + 1} = \sqrt[3]{8} = 2$$

Option (b):

$$= \sqrt{a + 2} = \sqrt{7 + 2} = \sqrt{9} = 3$$

Option (c):

$$= a - 3 = 7 - 3 = 4$$

Option (d):

$$= (a + 8) \div 3 = 15 \div 3 = 5$$

Video Solution:



Q22. Text Solution:

Total 2-wheelers passed from the toll plaza

though all the five lanes together = $12 + 5 + 7 + 10 + 13 = 47$

Total 4-wheelers passed from the toll plaza

though all the five lanes together = $6 + 4 + 5 + 5 + 8 = 28$

$$\text{Required difference} = 47 - 28 = 19$$

Video Solution:



**Q23. Text Solution:**

Difference between total 2 - wheelers and 4 - wheelers passed from lane S = $10 - 5$

$$x = 5$$

Total 4 - wheelers passed from lane Q + 1 = $4 + 1$

$$y = 5$$

Hence, $x = y = 5$

$$x + y = 5 + 5 = 10$$

$$x^2 + y^2 = 5^2 + 5^2 = 25 + 25 = 50$$

$$x - y = 5 - 5 = 0$$

Video Solution:**Q24. Text Solution:**

Total revenue collected from all the vehicles from lane R of the toll plaza

$$= (7 \times 180) + (5 \times 240) = 1260 + 1200 = \text{Rs.} 2460$$

Total revenue collected from all the vehicles from lane S of the toll plaza =

$$(10 \times 180) + (5 \times 240) = 1800 + 1200 =$$

Rs. 3000

$$\begin{aligned} \text{Required percent} &= \left(\frac{2460}{3000} \right) \times 100 \\ &= 82\% \end{aligned}$$

Video Solution:**Q25. Text Solution:**

Lanes	Total 2 - wheelers passed	Total 4 - wheelers passed
P	$16x = 12$	$8x = 6$
Q	$\frac{20x}{3} = 5$	$\frac{16x}{3} = 4$
R	$\frac{28x}{3} = 7$	$\frac{20x}{3} = 5$
S	$\frac{40x}{3} = 10$	$\frac{20x}{3} = 5$
T	$\frac{52x}{3} = 13$	$\frac{32x}{3} = 8$

Required Sum = $12 + 7 + 10 = 29$

Video Solution:**Q26. Text Solution:**

The number of AirAsia airplanes Departed from Mangalore = 20

The number of AirAsia airplanes Arrived at Bhopal = 18

To maximize the number of AirAsia airplanes that depart from Mangalore and reach Bhopal, we need to minimize the number of AirAsia airplanes from Mangalore to the other four cities i.e., Dehradun, Raipur, Ranchi, and Goa which is 1 airplane for each city.

Also, the number of AirAsia airplanes arrived at Bhopal from other four cities i.e., Dehradun, Raipur, Ranchi, and Goa which is 1 airplane for each city.

According to the question,

Therefore, the maximum number of AirAsia airplanes Departed from Mangalore to Bhopal = $20 - 4 = 16$

The maximum number of AirAsia airplanes that Arrived from Mangalore to Bhopal = $18 - 4 = 14$

Hence, the maximum number of AirAsia airplanes that departed from Mangalore and reached Bhopal = 14

Video Solution:

**Q27. Text Solution:**

The maximum number of Kingfisher airplanes that depart from Bhopal and can reach Raipur =
The number of Kingfisher airplanes departing from Bhopal – 4
 $= 32 - 4 = 28$.

In another way,

The maximum number of Kingfisher airplanes that can reach Raipur and depart from Bhopal =
The number of Kingfisher airplanes reaching Raipur – 4
 $= 13 - 4 = 9$.

Hence, the maximum number of AirAsia airplanes that departed from Bhopal and reached Raipur =
Minimum value of $[28, 9]$
 $= 9$

Video Solution:**Q28. Text Solution:**

For AirAsia:

The maximum number of AirAsia airplanes that depart from Goa and can reach Dehradun = The number of AirAsia airplanes departing from Goa – 4
 $= 24 - 4 = 20$.

In another way,

The maximum number of AirAsia airplanes that can reach Dehradun and depart from Goa = The number of AirAsia airplanes reaching Dehradun – 4
 $= 28 - 4 = 24$.

Hence, the maximum number of AirAsia airplanes that departed from Goa and reached Dehradun =
Minimum value of $[20, 24]$

$= 20$

For Indigo:

The maximum number of Indigo airplanes that departed from Goa and reached Dehradun =
Minimum value of $[(30 - 4), (31 - 4)]$
 $= \text{Minimum value of } [26, 27]$

$= 26$

For Kingfisher:

The maximum number of Kingfisher airplanes that departed from Goa and reached Dehradun =
Minimum value of $[(23 - 4), (34 - 4)]$
 $= \text{Minimum value of } [19, 30]$

$= 19$

For SpiceJet:

The maximum number of SpiceJet airplanes that departed from Goa and reached Dehradun =
Minimum value of $[(18 - 4), (12 - 4)]$
 $= \text{Minimum value of } [14, 8]$

$= 8$

Therefore, the maximum number of airplanes departed from Goa and reached Dehradun $= 20 + 26 + 19 + 8 = 73$.

Video Solution:**Q29. Text Solution:**

From Goa:

The maximum number of SpiceJet airplanes departing from Goa can reach any other city $= 18 - 4 = 14$

To:

Ranchi:

The maximum number of SpiceJet airplanes that departed from Goa and reached Ranchi =
Minimum value of $[14, (13 - 4)]$
 $= \text{Minimum value of } [14, 9]$

$= 9$

Raipur:

The maximum number of SpiceJet airplanes that departed from Goa and reached Raipur =
Minimum value of $[14, (18 - 4)]$
 $= \text{Minimum value of } [14, 14]$



= 14

Bhopal:

The maximum number of SpiceJet airplanes that departed from Goa and reached Bhopal =

Minimum value of $[14, (21 - 4)]$

= Minimum value of $[14, 17]$

= 14

Dehradun:

The maximum number of SpiceJet airplanes that departed from Goa and reached Dehradun =

Minimum value of $[14, (12 - 4)]$

= Minimum value of $[14, 8]$

= 8

Mangalore:

The maximum number of SpiceJet airplanes that departed from Goa and reached Mangalore =

Minimum value of $[14, (19 - 4)]$

= Minimum value of $[14, 15]$

= 14

From Goa, a maximum of 14 airplanes to Raipur, Bhopal, or Mangalore.

From Ranchi:

The maximum number of SpiceJet airplanes departing from Ranchi can reach any other city = $20 - 4 = 16$

To:

Goa:

The maximum number of SpiceJet airplanes that departed from Ranchi and reached Goa =

Minimum value of $[16, (13 - 4)]$

= Minimum value of $[16, 9]$

= 9

Raipur:

The maximum number of SpiceJet airplanes that departed from Ranchi and reached Raipur =

Minimum value of $[16, (18 - 4)]$

= Minimum value of $[16, 4]$

= 14

Bhopal:

The maximum number of SpiceJet airplanes that departed from Ranchi and reached Bhopal =

Minimum value of $[16, (21 - 4)]$

= Minimum value of $[16, 17]$

= 16

Dehradun:

The maximum number of SpiceJet airplanes that departed from Ranchi and reached Dehradun =

Minimum value of $[16, (12 - 4)]$

= Minimum value of $[16, 8]$

= 8

Mangalore:

The maximum number of SpiceJet airplanes that departed from Ranchi and reached Mangalore =

Minimum value of $[16, (19 - 4)]$

= Minimum value of $[16, 15]$

= 15

From Ranchi, a maximum of 16 airplanes to Bhopal.

From Raipur:

The maximum number of SpiceJet airplanes departing from Raipur can reach any other city = $13 - 4 = 9$

Already less than Ranchi to Bhopal.

From Bhopal:

The maximum number of SpiceJet airplanes departing from Bhopal can reach any other city = $21 - 4 = 17$

To:

Goa:

The maximum number of SpiceJet airplanes that departed from Bhopal and reached Goa =

Minimum value of $[17, (13 - 4)]$

= Minimum value of $[17, 9]$

= 9

Ranchi:

The maximum number of SpiceJet airplanes that departed from Bhopal and reached Ranchi =

Minimum value of $[17, (13 - 4)]$

= Minimum value of $[17, 9]$

= 9

Raipur:

The maximum number of SpiceJet airplanes that departed from Bhopal and reached Raipur =

Minimum value of $[17, (18 - 4)]$

= Minimum value of $[17, 14]$

= 14

Dehradun:

The maximum number of SpiceJet airplanes that departed from Bhopal and reached Dehradun =



Minimum value of $[17, (12 - 4)]$

= Minimum value of $[17, 8]$

= 8

Mangalore:

The maximum number of SpiceJet airplanes that departed from Bhopal and reached Mangalore =

Minimum value of $[17, (19 - 4)]$

= Minimum value of $[17, 15]$

= 15

From Bhopal, a maximum of 15 airplanes to Mangalore.

From Dehradun:

The maximum number of SpiceJet airplanes departing from Dehradun can reach any other city = $14 - 4 = 10$

Already less than Ranchi to Bhopal.

Hence, a maximum of 16 SpiceJet airplanes departed from any given city to any other five cities i.e. from Ranchi to Bhopal.

Video Solution:



Q30. Text Solution:

For AirAsia:

The maximum number of AirAsia airplanes that depart from Raipur and can reach Goa = AirAsia airplanes departing from Raipur - 4

= $20 - 4$

= 16

In another way,

The maximum number of AirAsia airplanes to Goa from Raipur = AirAsia airplanes reached Goa - 4

= $28 - 4$

= 24

Hence, the maximum number of AirAsia airplanes that departed from Raipur and reached Goa =

Minimum value of $[16, 24]$

= 16

For Indigo:

The maximum number of Indigo airplanes that depart from Raipur and can reach Goa = Indigo airplanes departing from Raipur - 4

= $33 - 4$

= 29

In another way,

The maximum number of Indigo airplanes to Goa from Raipur = Indigo airplanes reached Goa - 4

= $28 - 4$

= 24

Hence, the maximum number of Indigo airplanes that departed from Raipur and reached Goa =

Minimum value of $[29, 24]$

= 24

Therefore, the maximum number of AirAsia and Indigo airplanes that departed from Raipur and reached Goa = $16 + 24$

= 40

Video Solution:



[Android App](#) | [iOS App](#) | [PW Website](#)